

ICDFA GRC102: Information Security Governance

Week 4 Linux Lab Assignment: Implementing Security Monitoring and Auditing

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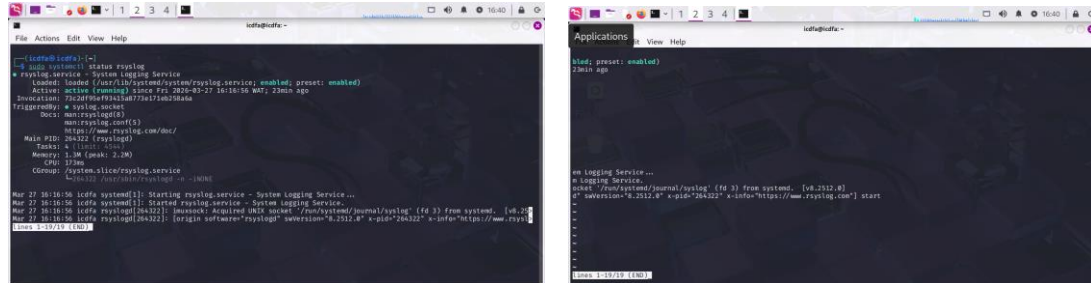
Part 1: Configuring Linux Logging and Auditing

Deliverables:

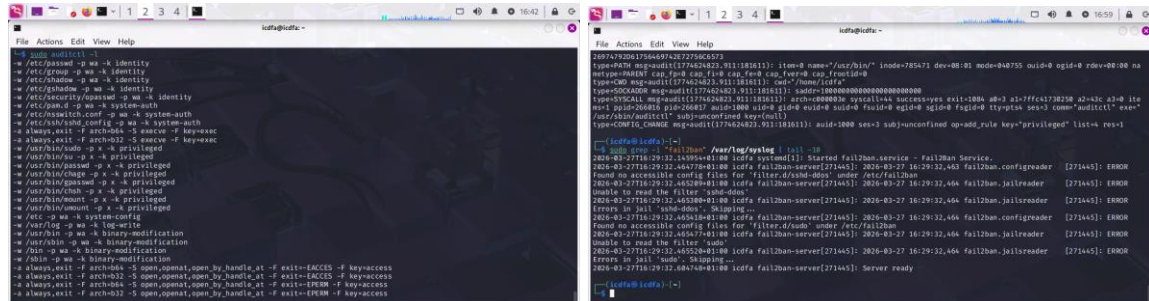
- Screenshots showing the configuration of rsyslog, auditd, and fail2ban

ANSWER

1. Fig 1 & 2 showing rsyslog configuration



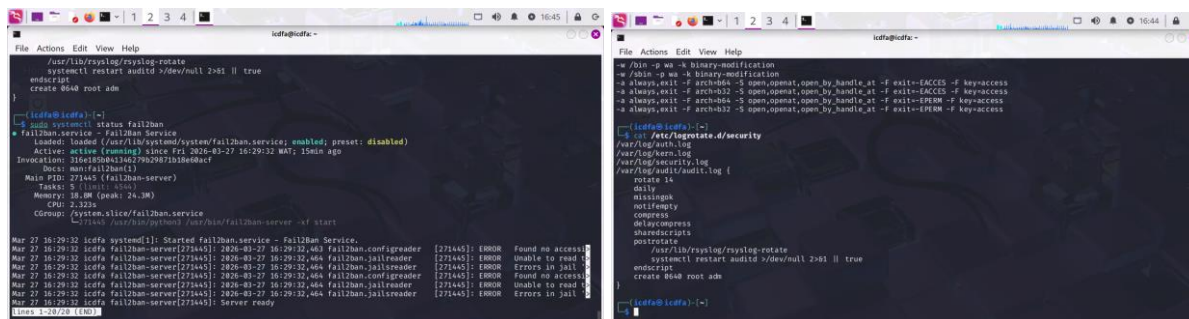
2. Fig 3 & 4 showing auditd configuration & fail2ban



- Output from the test script showing the logging and auditing in action

ANSWER

3. Fig 5 & 6 showing logging and auditing in action



- A brief explanation (1-2 paragraphs) of how these logging and auditing mechanisms support security governance

ANSWER:

Security logging and auditing are essential components of effective security governance, as they provide visibility into system activities and potential security incidents. In this lab, rsyslog was configured to centralize and store critical logs such as authentication attempts, kernel messages, and security-related events. Auditd was used to monitor sensitive files, system calls, and privileged commands, ensuring that any unauthorized access or changes are recorded. Additionally, Fail2ban was implemented to detect and respond to repeated authentication failures by automatically banning suspicious IP addresses.

These mechanisms enhance security governance by enabling continuous monitoring, incident detection, and accountability. Logs generated from these tools support forensic investigations, compliance requirements, and proactive threat detection. By integrating logging, auditing, and automated response, organizations can significantly improve their ability to detect, analyze, and respond to security threats in a timely and efficient manner.

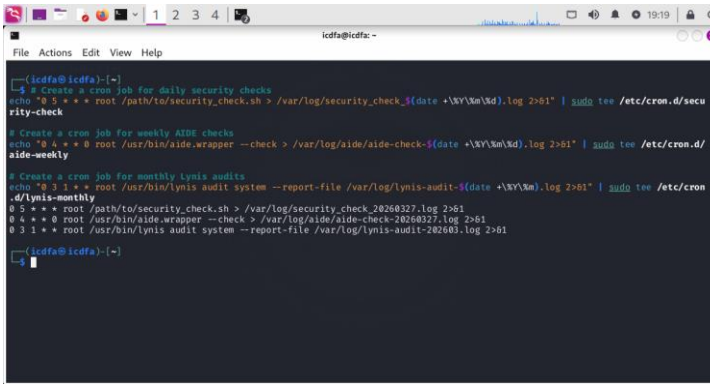
Part 2: Implementing Security Monitoring Tools

Deliverables:

- Screenshots showing the installation and configuration of AIDE, Lynis, and Logwatch

ANSWER

1. *Figure 2.1 showing configuration of AIDE*

A terminal window titled 'icdfa@icdfa: ~' showing a series of commands to set up security monitoring. The commands include creating cron jobs for daily security checks, weekly AIDE checks, and monthly Lynis audits. The output shows the cron jobs being added to the system.

```
icdfa@icdfa:~$ cat <>
#!/bin/bash
# Create a cron job for daily security checks
echo "0 5 * * * root /path/to/security_check.sh > /var/log/security_check_$(date +%Y%m%d).log 2>&1" | sudo tee /etc/cron.d/security-check

# Create a cron job for weekly AIDE checks
echo "0 4 * * 0 root /usr/bin/aide.wrapper --check > /var/log/aide/aide-check-$(date +%Y%m%d).log 2>&1" | sudo tee /etc/cron.d/aide-weekly

# Create a cron job for monthly Lynis audits
echo "0 3 1 * * root /usr/bin/lynis audit system --report-file /var/log/lynis-audit-$(date +%Y%m).log 2>&1" | sudo tee /etc/cron.d/lynis-monthly

cat <>
0 5 * * * root /path/to/security_check.sh > /var/log/security_check_20260327.log 2>&1
0 4 * * 0 root /usr/bin/aide.wrapper --check > /var/log/aide/aide-check-20260327.log 2>&1
0 3 1 * * root /usr/bin/lynis audit system --report-file /var/log/lynis-audit-202603.log 2>&1

icdfa@icdfa:~$
```

- The Lynis audit report

ANSWER

- A brief explanation (1-2 paragraphs) of how these security monitoring tools enhance the security governance program

ANSWER

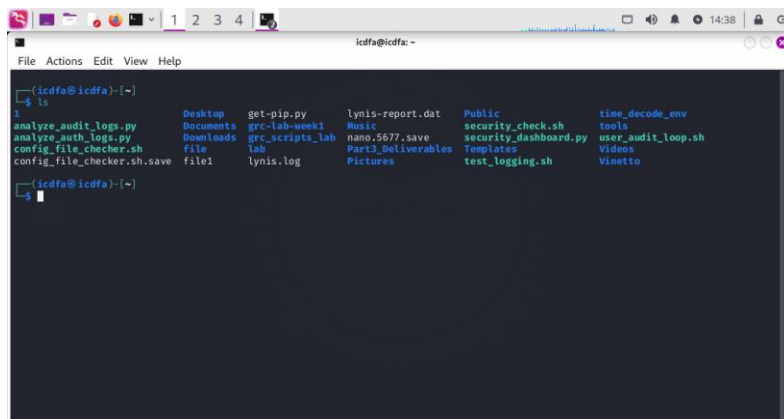
Security monitoring tools such as AIDE, Lynis, and Logwatch significantly enhance an organization's security governance by providing continuous visibility into system integrity, vulnerabilities, and operational activities. AIDE ensures file integrity by detecting unauthorized changes to critical system files, which is essential for identifying potential compromises. Lynis performs comprehensive security audits, identifying misconfigurations, vulnerabilities, and areas for system hardening. Logwatch complements these tools by analyzing system logs and generating structured reports that simplify monitoring and incident detection.

Additionally, the implementation of automated scripts and scheduled tasks using cron ensures continuous and consistent security checks without manual intervention. This improves efficiency, reduces human error, and enables proactive threat detection. Together, these tools support compliance, strengthen audit trails, and enhance the overall security posture by enabling timely detection, analysis, and response to security events.

3. Deliverables:

- The auth_analysis_report.txt file generated by your script

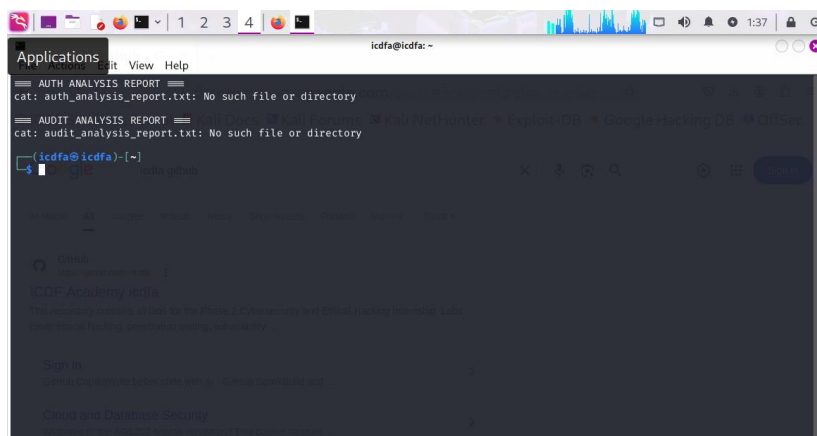
ANSWER



```
icdfa@icdfa: ~  
File Actions Edit View Help  
icdfa@icdfa:~$ ls  
analyze_audit_logs.py  Desktop  get-pip.py  lynis-report.dat  Public  time_decode_env  
analyze_auth_logs.py  Documents  grc-lab-week1  Music  security_check.sh  tools  
config_file_checker.sh  Downloads  grc_scripts_lab  nano.5677.save  security_dashboard.py  user_audit_loop.sh  
config_file_checker.sh.save  file  lab  Part3_Deliverables  Templates  Videos  
file1  lynis.log  Pictures  test_logging.sh  Vinetto  
icdfa@icdfa:~$
```

- The audit_analysis_report.txt file generated by your script

ANSWER

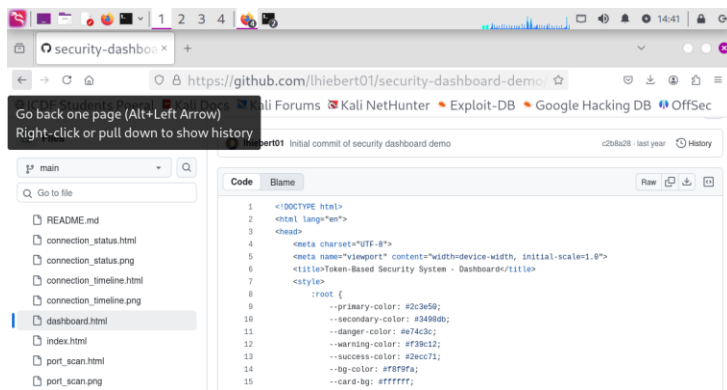


```
icdfa@icdfa: ~  
Applications View Help  
icdfa@icdfa:~$ cat: auth_analysis_report.txt: No such file or directory  
icdfa@icdfa:~$ cat: audit_analysis_report.txt: No such file or directory  
icdfa@icdfa:~$
```

- The security_dashboard.html file

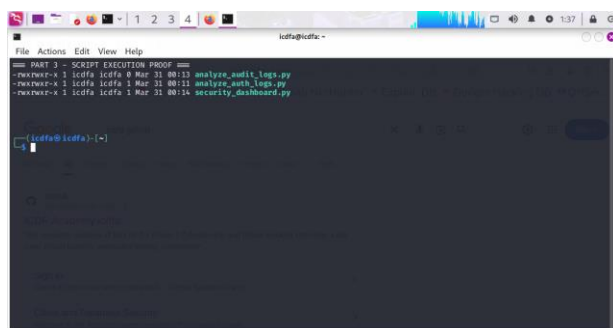
ANSWER

<https://github.com/lhiebert01/security-dashboard-demo/blob/main/dashboard.html>



- Screenshots of the visualizations generated by your scripts

ANSWER



- A brief explanation (1-2 paragraphs) of how these analysis and visualization tools enhance security monitoring and auditing

ANSWER

Authentication and audit log analysis tools improve security monitoring by providing visibility into system activities and potential threats. By analyzing authentication logs, security teams can identify suspicious login attempts, brute force attacks, and unauthorized access patterns.

Visualization of this data makes it easier to detect trends, such as repeated failed login attempts from specific IP addresses.

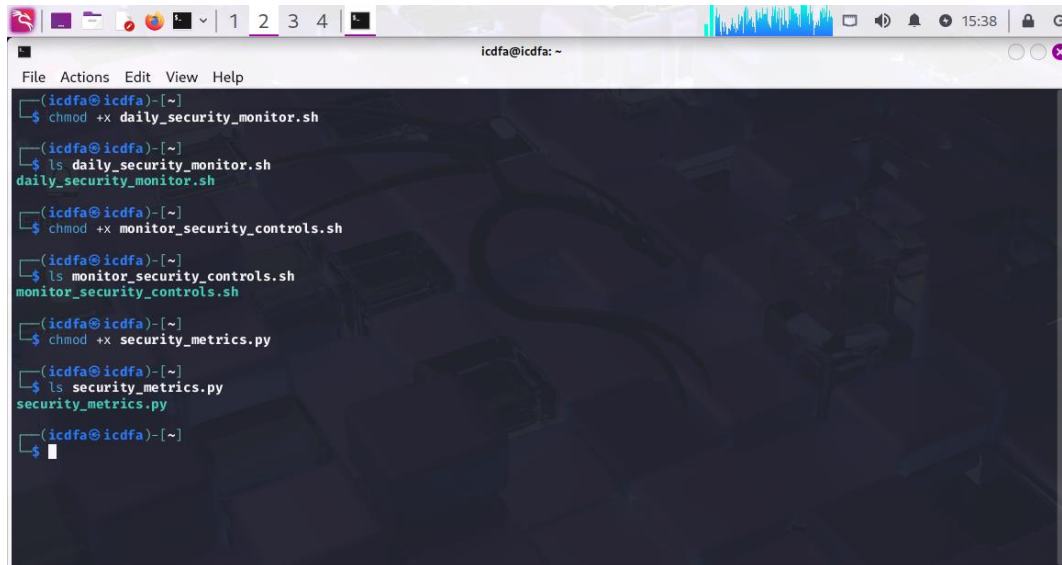
Additionally, audit log analysis helps track system-level activities, including command executions and user actions. The dashboard integrates multiple data sources into a single view, enabling faster decision-making and incident response. Overall, these tools enhance proactive security monitoring, improve threat detection, and support compliance with security policies.

PART 4

Deliverables:

- The security_report.txt file generated by the daily monitoring script

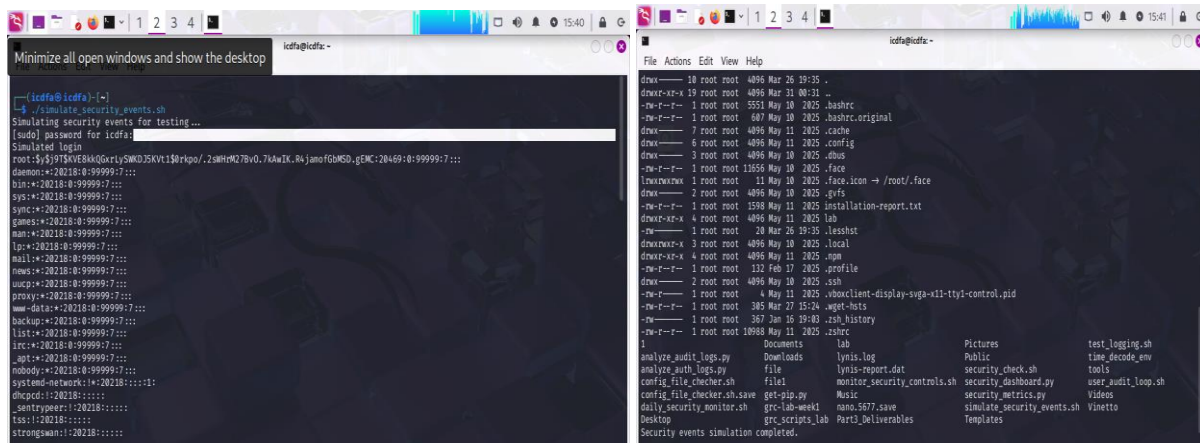
ANSWER



```
icdfa@icdfa: ~  
File Actions Edit View Help  
icdfa@icdfa:~$ chmod +x daily_security_monitor.sh  
icdfa@icdfa:~$ ls daily_security_monitor.sh  
daily_security_monitor.sh  
icdfa@icdfa:~$ chmod +x monitor_security_controls.sh  
icdfa@icdfa:~$ ls monitor_security_controls.sh  
monitor_security_controls.sh  
icdfa@icdfa:~$ chmod +x security_metrics.py  
icdfa@icdfa:~$ ls security_metrics.py  
security_metrics.py  
icdfa@icdfa:~$
```

- The control_changes.txt file generated by the control monitoring script

ANSWER

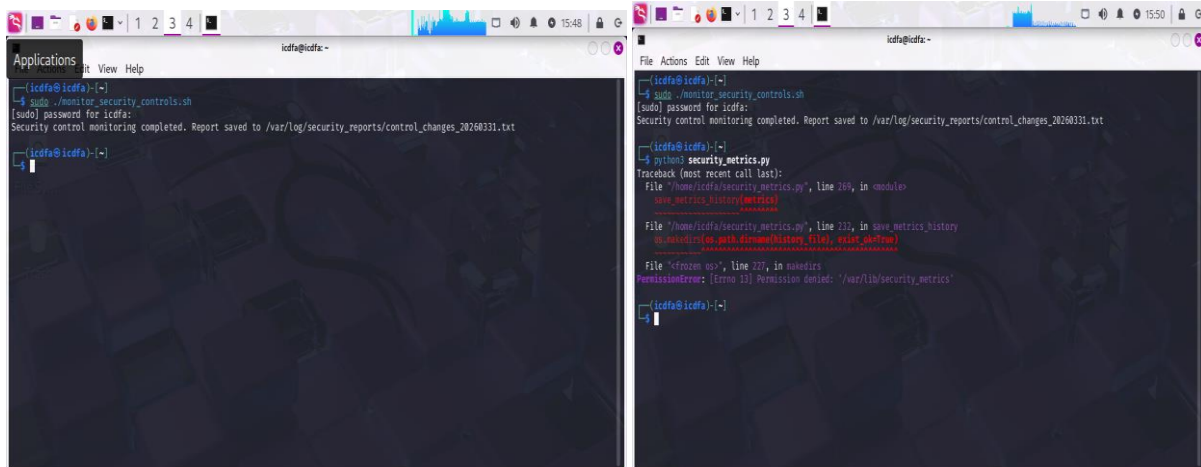


```
icdfa@icdfa:~$ ./simulate_security_events.sh  
Simulating security events for testing...  
Simulated login  
root:Sy$1915KVE8KQGrly5W0J5KV11$0rkpo/.2mHm278v0.7k6wIX_R4janofoGwSD_gENC:2046910:99999:7:::  
daemon:*20218:0:99999:7:::  
bin:*20218:0:99999:7:::  
sys:*20218:0:99999:7:::  
sync:*20218:0:99999:7:::  
games:*20218:0:99999:7:::  
man:*20218:0:99999:7:::  
lp:*20218:0:99999:7:::  
mail:*20218:0:99999:7:::  
news:*20218:0:99999:7:::  
uucp:*20218:0:99999:7:::  
proxy:*20218:0:99999:7:::  
www-data:*20218:0:99999:7:::  
backup:*20218:0:99999:7:::  
list:*20218:0:99999:7:::  
irc:*20218:0:99999:7:::  
_apt:*20218:0:99999:7:::  
nobody:*20218:0:99999:7:::  
systemd-network:*20218:0:99999:7:::  
dhcpcd:*20218:0:99999:7:::  
_systemd:*20218:0:99999:7:::  
_systemd:*20218:0:99999:7:::  
strongswan:*20218:0:99999:7:::  
Security events simulation completed.
```

```
icdfa@icdfa:~$ ls  
Documents  Downloads  lab  Pictures  test_logging.sh  
analyze_audit_logs.py  file  lynis.log  Public  time_decode_env  
analyze_auth_logs.py  file  lynis-report.dat  security_check.sh  tools  
config_file_checker.sh  file  monitor_security_controls.sh  security_dashboard.py  user_audit_loop.sh  
config_file_checker.sh.save  get-pip.py  Music  security_metrics.py  Videos  
daily_security_monitor.sh  grc-lab-week1  nano.5677.save  simulate_security_events.sh  Vinetto  
Desktop  grc_scripts_lab  Part3_Deliverables  Templates
```

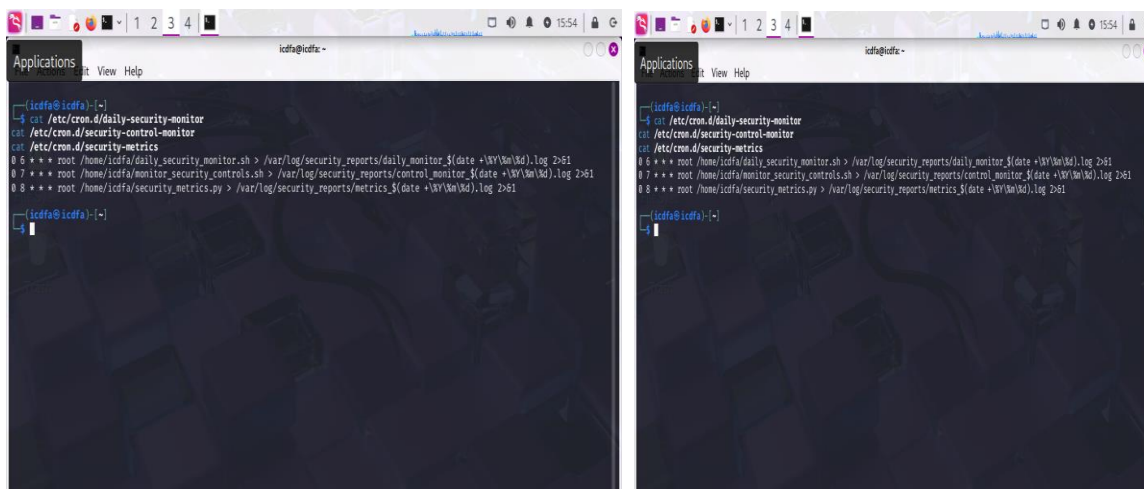
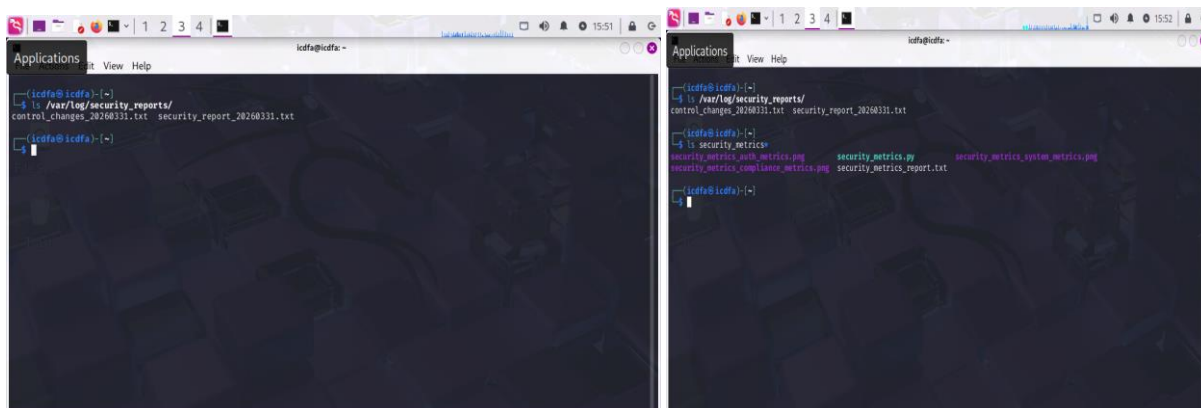
- The security_metrics_report.txt file and associated visualizations

ANSWER



- Screenshots showing the scheduled cron jobs

ANSWER



- A brief explanation (1-2 paragraphs) of how continuous monitoring supports security governance

ANSWER

Continuous monitoring is a critical component of security governance as it ensures that systems are regularly assessed for vulnerabilities, unauthorized changes, and suspicious activities. By automating daily security checks, organizations can detect threats early, maintain compliance with security policies, and respond quickly to incidents.

The implementation of automated scripts and scheduled cron jobs enables real-time visibility into system security posture, reduces manual effort, and ensures consistency in monitoring. This approach supports proactive risk management and strengthens overall cybersecurity resilience.